

- (c) Electric separation (d) Reverse osmosis
U.P.P.C.S. (Mains) 2013

Ans. (d)

See the explanation of above question.

27. By which process, Alum cleans the muddy water?

- (a) Absorption (b) Adsorption
(c) Coagulation (d) Dialysis

R.A.S./R.T.S.(Pre) 2000

Ans. (c)

Alum (aluminium sulphate) is added to the water to destabilize natural fine particulate matter suspended in the water. This process is known as coagulation. The impurities found in water suspensions consists of charged colloids ranging in size from 5mm to 1mm and particles greater than 0.5 mm. The addition of alum causes these particles to collide and clump together to form heavy cluster which will settle in water.

28. The gas used in the manufacture of Vanaspati Ghee from Vanaspati Oil is –

- (a) Hydrogen (b) Oxygen
(c) Nitrogen (d) Carbon dioxide

60th to 62nd B.P.S.C. (Pre) 2016
U.P.P.C.S. (Mains) 2013

Ans. (a)

Hydrogen gas is used in the manufacturing of vanaspati ghee. To prepare the vanaspati ghee, hydrogen gas is passed through vegetable oils under 8-10 atmospheric pressure in the presence of nickel (Ni) powder (which works as catalyst) at a temperature of 150-180°C, and oil changes into vegetable ghee or solid fat known as vanaspati ghee. This process is called hydrogenation. Hydrogenation is a process in which unsaturated compounds combine with hydrogen in the presence of a catalyst and in other suitable conditions and are converted into saturated compounds.

29. Which one of the following catalysts is used in hydrogenation of vegetable oils?

- (a) Zinc (b) Platinum
(c) Nickel (d) Iron

U.P. P.C.S. (Mains) 2016

Ans. (c)

See the explanation of above question.

30. Catalyst used in hydrogenation of vegetable oil into vanaspati ghee is :

- (a) Pd (b) Fe
(c) Ni (d) None of the above

70th B.P.S.C. (Pre) (Re-Exam) 2024

Ans. (c)

See the explanation of above question.

C. Sulphur, Nitrogen, Halogen, Inert Gases

Notes

Sulphur (S) :

- It is a nonmetallic, abundant and multivalent element. Elemental sulphur at room temperature is a bright yellow, crystalline solid.
- The elemental sulphur is used in matches, insecticides and fungicides.
- The main commercial use of sulphur is the production of sulphuric acid for sulphate and phosphate fertilizers.
- It is one of the core elements needed for biochemical functioning and is an elemental macronutrient for all living organisms.
- It is also called 'brimstone' (burning stone).

Sulphur dioxide (SO₂) :

- It is a toxic gas with a burnt match smell.
- It is released naturally by volcanic activities and is produced as a by-product of the burning of fossil fuels contaminated with sulphur compounds.
- It is used as a precursor to sulfuric acid, as a preservative for dried fruits, in winemaking, as a refrigerant, etc.
- SO₂ is also a good reducing agent. In the presence of water, it is able to decolorize substances.
- It is a major air pollutant and has significant impacts on human health.

Sulphur Hexafluoride (SF₆) :

- It is an inorganic, colourless, odourless, non-flammable, extremely potent greenhouse gas and an excellent electrical insulator.
- It has an octahedral geometry, consisting of six fluorine atoms attached to a central sulphur atom. It is a hypervalent compound.
- It is used as a gaseous dielectric medium in the electric industry. Other main usages include as inert gas for the casting of magnesium and as an inert filling for insulated glazing windows.

Nitrogen (N) :

- Nitrogen gas (N₂) forms about 78% of the Earth's atmosphere.
- It is a naturally occurring element that is essential for growth and reproduction in both plants and animals. It

is found in amino acids that make up proteins, in nucleic acids that comprise the heredity material and life blueprint for all cells and in many organic and inorganic compounds.

- Aircrafts tyres are filled with nitrogen because nitrogen gas is mostly inert, meaning that it requires more energy to react with other substances.
- In relation to this, another reason is that nitrogen is less likely to diffuse through the rubber than oxygen, maintaining tyre pressure stable over the long term.
- Insectivorous plants have adapted to grow in places where the soil is poor in nutrients especially nitrogen such as acidic bogs. These plants get nitrogen from insects.
- Approximately 440 species of insectivorous plants have been discovered. In India, such plants are found in Darjiling, Nainital, Kashmir, etc.

Nitrous Oxide (N₂O) :

- It is commonly known as **laughing gas**.
- Its formula is N₂O.
- At room temperature, it is colourless, non-flammable gas, with a slight metallic scent and taste and soluble in water.
- On the increasing temperature, it is a powerful oxidiser similar to molecular oxygen.
- It is used in surgery and dentistry, for its **anaesthetic** and pain-reducing effects.

Ammonia (NH₃) :

- It is a compound of nitrogen and hydrogen.
- It is a colourless gas with a specific pungent smell.
- It is a common nitrogenous waste and it contributes significantly to the nutritional needs of organisms by serving as a precursor to food and fertilizers.
- Ammonia mainly used as fertilizers either as its salts, solutions or anhydrously.
- Its solution in water is used as a general-purpose cleaner for many surfaces.

Tear Gas :

- It is formally known as lacrymatory agent.
- It is a chemical weapon that causes severe eye and respiratory pain, skin irritation, bleeding and even blindness.
- In the eye, it stimulates the nerves of the lacrimal gland to produce tears.
- Some main tear gases are Alphachloroacetophenone, 2-Chlorobenzalmalononitrile etc.
- Chlorine is also used in preparation of tear gas.

Halogens :

- These are a group of elements in the periodic table.

- These have been kept in group 17 of the periodic table.
- Its members are fluorine, chlorine, bromine, iodine and astatine.
- The artificially created element **Tennessine (Ts)** with atomic number 117 may also be a halogen.
- Fluorine and chlorine, are gases. While iodine is a blue-black opaque crystalline solid and bromine is liquid at normal temperature and pressure. Astatine is a radioactive element.
- Halogens are the most reactive non-metallic elements.
- Fluorine is the most reactive element among halogens.
- Compounds of bromine are used as analgesics.

Inert Gases (Noble Gases) :

- An inert gas or noble gas is a gas that has extremely low reactivity with other substances. These gases do not undergo chemical reactions under certain given conditions.
- The noble gases are - helium, neon, argon, krypton, xenon, radon and may be newly found element with atomic number 118–**Oganesson (Og)**.
- These are kept in group 18 of the periodic table.
- Their outer shell of valence electrons is considered to be full, giving them little tendency to participate in chemical reactions.
- Helium is used in breathing gas by deep-sea divers to prevent oxygen, nitrogen and carbon dioxide toxicity.
- Argon is used in incandescent lamps to prevent the hot tungsten filament from oxidizing.

Xenon (Xe) :

- It is a chemical element of zero group.
- It is inert gas with atomic number 54.
- It is a colourless, dense, odourless noble gas found in the Earth's atmosphere in trace amounts.
- It is used in photographic flashes, in high-pressure arc lamps for motion picture projection and in high- pressure arc lamps to produce ultra-violet light. It is used in instruments for radiation detection e.g. neutron and X-ray counters and bubble chambers.
- It is also used as a general anaesthetic and as the propellant for ion thrusters in spacecraft.
- It was discovered by William Ramsay and Morris Travers in 1898.
- It is heaviest than air and also known as Stranger gas.

Heliox :

- It is a breathing gas composed of a mixture of helium (He) and oxygen (O). The main reason for this breathing mix is to reduce the proportion of nitrogen below those of air, to allow the gas mixture to be breathed safely on deep dives. A lower proportion of nitrogen is required to reduce nitrogen narcosis. Heliox generates less airway resistance than air and thereby required less mechanical energy to ventilate the lungs.

Question Bank**1. Which gas is known as 'noble gas'?**

- (a) Hydrogen (b) Oxygen
(c) Helium (d) Carbon dioxide

M.P. P.C.S. (Pre) 2016**Ans. (c)**

The noble gases make a group of chemical elements with similar properties under standard conditions. They are all odourless, colourless, monatomic gases with very low chemical reactivity. These gases do not undergo chemical reactions under certain given conditions. The six noble gases that occur naturally are Helium (He), Neon (Ne), Argon (Ar), Krypton (Kr), Xenon (Xe) and the radioactive Radon (Rn).

2. Which of the following elements is not a Noble gas?

- (a) Argon (b) Krypton
(c) Actinium (d) Radon

U.P. P.C.S. (Pre) 2023**Ans. (c)**

See the explanation of above question.

3. Percentage amount (by volume) of which one of the following gas is lowest in the atmosphere?

- (a) Argon (b) Carbon dioxide
(c) Nitrogen (d) Oxygen

R.A.S./R.T.S.(Pre) 2007**Ans. (b)**

The atmosphere contains many gases mostly in a small amount including some pollutants and greenhouse gases. The most abundant gas in the atmosphere (by volume) is nitrogen (about 78%) with oxygen second at almost 20.95%. Argon (about 0.93%), an inert gas is the third most abundant gas in the atmosphere. While carbon dioxide percentage amount in the atmosphere is about 0.03%.

4. What is the approximate percentage of Nitrogen present in the atmosphere?

- (a) 10-11% (b) 18-20%

(c) 40-42%

(d) 78-79%

M.P.P.C.S. (Pre) 2006**Ans. (d)**

See the explanation of above question.

5. Which gas is not in the gaseous cycle ?

- (a) N₂ (b) O₂
(c) Carbon (d) H₂
(e) None of these

Chhattisgarh P.C.S. (Pre) 2014**Ans. (d)**

Biogeochemical cycles can be classed as gaseous in which the reservoir is the air or the oceans (via evaporation), and sedimentary in which the reservoir is earth's crust. Gaseous cycles include Nitrogen, Oxygen, Carbon and Water. Sedimentary cycles include Iron, Calcium, Phosphorus, Sulphur and other more earthbound elements.

6. Which of the following pair of gases is used for the respiratory activities of divers?

- (a) Oxygen and Helium (b) Oxygen and Neon
(c) Oxygen and Nitrogen (d) Oxygen and Argon

U.P. R.O./A.R.O. (Mains) 2016**U.P. P.C.S. (Mains) 2014****Ans. (a)**

The deep sea divers carry cylinders which contain a mixture of oxygen and helium. The presence of helium generates less airway resistance than air and thereby requires less mechanical energy to ventilate the lungs. Helium is much less soluble in water than many other gases, such as nitrogen. The low solubility means it does not enter the blood stream even under pressure.

7. Which gas is mixed with Oxygen for respiration by divers in deep sea?

- (a) Helium (b) Nitrogen
(c) Xenon (d) Ammonia

R.A.S./R.T.S.(Pre) 2007**Ans. (a)**

See the explanation of above question.

8. Which one of the following mixture of gas is used by the deep sea divers?

- (a) Mixture of Helium – Oxygen
(b) Mixture of Helium – Nitrogen
(c) Mixture of Oxygen – Acetylene
(d) Mixture of Oxygen – Hydrogen

U.P. Lower Sub. (Pre) 2004**Ans. (a)**

See the explanation of above question.

9. What is the shape of sulphur hexafluoride molecule?

- (a) Trigonal pyramid (b) Octahedral
(c) Planar (d) Tetrahedral

43rd B.P.S.C. (Pre) 1999

Ans. (b)

Sulphur hexafluoride (SF_6) is an inorganic, colourless, odourless and non-flammable compound. SF_6 molecule has an octahedral shape, consisting of six fluorine atoms attached to a central sulphur atom.

10. Easily soluble in water –

- (a) Carbon (b) Nitrogen
(c) Ammonia (d) Iodine

U.P.P.C.S. (Pre) 1993

Ans. (c)

Ammonia being a polar molecule dissolves readily in water. This is due to the hydrogen atoms of ammonia which are bonded with a highly electronegative nitrogen and the hydrogen atoms of water molecules which are bonded with the highly electronegative oxygen atom. Ammonia and water are both polar molecules and they are both capable of hydrogen bonding, so they are highly miscible.

11. Which is used as Laughing gas –

- (a) Nitrous oxide (b) Nitrogen dioxide
(c) Nitrogen trioxide (d) Nitrogen tetra oxide

U.P.P.C.S. (Mains) 2011

U.P.U.D.A./L.D.A. (Mains) 2010

40th B.P.S.C. (Pre) 1995

Ans. (a)

Nitrous oxide (N_2O) is also known as laughing gas. It is a colourless gas with a slight metallic odour and taste. Its inhalation leads to disorientation, euphoria, numbness, loss of coordination, dizziness and ultimately a loss of consciousness. It is also used as the anaesthetic gas. This gas is used in medical and dental procedures as a sedative. It helps to relieve anxiety before the procedure and allow the patient to relax.

12. Which of the following is known as laughing gas?

- (a) Nitrous oxide (b) Chlorofluorocarbon
(c) Carbon monoxide (d) Ammonia

U.P. P.C.S. (Pre) 2023

Ans. (a)

See the explanation of above question.

13. The 'laughing gas' is :

- (a) Hydrogen peroxide (b) Nitrous oxide

- (c) Carbon monoxide (d) Sulphur dioxide

M.P.P.C.S. (Pre) 2017

Ans. (b)

See the explanation of above question.

14. Which of the following is also known as laughing gas?

- (a) Nitric oxide
(b) Nitrous oxide
(c) Nitrogen pentoxide
(d) Nitrogen
(e) None of the above/More than one of the above

65th B.P.S.C. (Pre) 2019

Ans. (b)

See the explanation of above question.

15. Laughing gas used as anaesthesia by doctors is –

- (a) Nitrogen (b) Nitrogen oxide
(c) Nitrous oxide (d) Nitrogen dioxide

44th B.P.S.C. (Pre) 2000

Ans. (c)

See the explanation of above question.

16. Which one of the following is also called Stranger Gas?

- (a) Argon (b) Neon
(c) Xenon (d) Nitrous Oxide

I.A.S. (Pre) 2008

Ans. (c)

Xenon is a gaseous element with symbol Xe and atomic number 54. It is a rare, odourless, colourless and tasteless gas. Xenon gas is also known as Stranger Gas. It belongs to the noble gas group and although generally unreactive, it can undergo a few chemical reactions. Its volume is very low in the atmosphere (about 0.08 parts per million). It was discovered in England by the Scottish chemist William Ramsay and British chemist Morris Travers in 1898.

17. Which one of the following is not correct?

- (a) Bordeaux mixture is a mixture of copper (II) sulphate and calcium hydroxide in water and is used as a fungicide.
(b) Boric acid is used in pharmaceutical industry and also as a food preservative.
(c) Dry ice is solid carbon dioxide
(d) Nitric oxide is laughing gas.

U.P.P.C.S. (Mains) 2010

Ans. (d)

Nitric oxide (NO) is not a laughing gas. In fact, nitrous oxide (N_2O) is known as laughing gas. Rest of the options are correct.

18. Discharge of nitrogen causes –

- (a) Increase in the amount of atmospheric nitrogen
- (b) Increase in the amount of nitrogen in lithosphere
- (c) The amount of nitrogen in lithosphere and in atmosphere unchanged
- (d) Destruction of nitrifying bacteria

U.P.P.C.S. (Mains) 2008

Ans. (c)

The nitrogen cycle is a process by which nitrogen is converted into its various chemical forms. This transformation can be carried out through both biological and physical processes. The nitrogen cycle includes fixation, ammonification, nitrification and denitrification. Denitrifying bacteria converts nitrate and nitrite ions into nitrogen. From discharge of this nitrogen, its amount in the lithosphere and in atmosphere remains unchanged.

19. Which one of the following is the permanent gas in the atmosphere?

- (a) Carbon dioxide
- (b) Ozone
- (c) Nitrogen
- (d) Neon

U.P. P.C.S. (Mains) 2017

Ans. (c)

Permanent Gases are those gases which are believed to be incapable of liquefaction and remains gaseous under normal conditions. These gases include hydrogen, nitrogen, carbon monoxide etc. Nitrogen is a permanent gas in the atmosphere. It accounts for about 78% of the atmosphere.

20. The gas used to inflate the tyres of an aircraft is –

- (a) Hydrogen
- (b) Nitrogen
- (c) Helium
- (d) Neon

U.P.R.O./A.R.O. (Pre) 2014

Uttarakhand U.D.A./L.D.A. (Pre) 2007

Ans. (b)

Nitrogen gas is used in the tyres of an aeroplane. This is because the nitrogen gas does not support combustion and can assist in preventing wheel fire when the aircraft lands (braking and high speed can produce dangerously high temperatures) unlike in oxygen. There are other benefits but effectively it is being the lowest cost gas that does not support combustion.

21. In order to compensate the deficiency of which of the following elements, the insectivorous plants trap and eat insects ?

- (a) Manganese
- (b) Nitrogen

(c) Magnesium

(d) Sulphur

U.P.R.O./A.R.O. (Mains) 2013

Ans. (b)

Carnivorous or insectivorous plant can trap and digest insects and other small animals. They do this to obtain vital nitrogen that they need to grow. Most of the plants absorb enough nitrogen from nitrates in the soil.

22. The soil in which insectivorous plants grow is deficient in :

- (a) Magnesium
- (b) Calcium
- (c) Nitrogen
- (d) Water

U.P. P.C.S. (Mains) 2016

Ans. (c)

Insectivorous plants cannot take their nitrogen from soil as they grow in waterlogged swampy soils deficient in nitrogenous compounds. Such plants obtain their nitrogen from small insects. For this work, these plants have evolved special mode of nutrition. Example of insectivorous plants are Nepenthes, Dionaea, Drosera, Utricularia, etc.

23. Which gas is usually filled in the balloon?

- (a) Hydrogen
- (b) Oxygen
- (c) Carbon dioxide
- (d) Helium

M.P. P.C.S. (Pre) 1991

Ans. (d)

A gas balloon flies in the air because it is filled with a gas which is lighter than air. Helium is mostly used in gas balloons. It is also inert in nature and does not form an explosive mixture with air.

24. Helium is preferred to Hydrogen in air balloons because it –

- (a) Is cheaper
- (b) Is less dense
- (c) Has greater lifting power
- (d) Does not form an explosive mixture with air

U.P.P.C.S. (Pre) 2011

Ans. (d)

Helium is preferred over hydrogen in air balloons because it is inert in nature. Also, hydrogen is highly flammable and explosive so it would make it unsafe to use in balloons.

25. Tear gas is :

- (a) Ammonia
- (b) Chlorine
- (c) Hydrogen Carbide
- (d) Hydrogen Sulphide

Uttarakhand P.C.S. (Pre) 2002

Ans. (b)

Tear gas is a chemical weapon that causes severe eye and respiratory pain, skin burning, vomiting and even blindness. In the eye, it stimulates the nerves of the lacrimal gland to produce tears. It is mostly used by the police force in controlling riots. Alphachloroacetophenone, 2-chlorobenzalmalononitrile and acrolein are generally used as tear gas. Chlorine (Cl_2) is also used in preparation of tear gas.

26. Which of the following gas is used as tear gas –

- (a) H_2 (b) SO_2
(c) NH_3 (d) Cl_2

43rd B.P.S.C. (Pre) 1999

Ans. (d)

See the explanation of above question.

27. The foul smell of dead fish is due to the compound

- (a) Amino compounds
(b) Aldehydic compounds
(c) Sulphur compounds
(d) Nitro compounds

Jharkhand P.C.S. (Pre) 2016

Ans. (a)

Trimethylamine is a tertiary amine that is structurally an amino compound in which every hydrogen atom of ammonia is replaced by a methyl group. The odour (smell) of dead fish is due to the presence of these compounds.

28. Which of the following remains in a solid state at normal temperature?

- (a) Chlorine (b) Bromine
(c) Iodine (d) Fluorine

Jharkhand P.C.S. (Pre) 2003

Ans. (c)

The Group Seven (7) elements are also known as the halogens. They include fluorine, chlorine, bromine and iodine which all have 7 electrons in their outer shell. Iodine is a bluish-black solid with a metallic lustre, thus it remains in a solid state at normal temperature.

29. The most reactive among the halogens is –

- (a) Fluorine
(b) Chlorine
(c) Bromine
(d) Iodine

I.A.S. (Pre) 1997

Ans. (a)

The halogens (Fluorine, Chlorine, Bromine, Iodine, Astatine) are non-metal elements which are highly electronegative and reactive. They constitute Group 17 of the periodic table. The newly created element Tennessine (Ts) with atomic number 117 may also be a halogen, but its chemical properties are still unknown. Fluorine is the smallest atom among the halogen elements. It also has the strongest ability to attract electrons from other elements making it the most reactive of the halogens.

30. Bromine is a :

- (a) Colourless gas
(b) Brown solid
(c) Highly inflammable gas
(d) Red liquid
(e) None of the above/More than one of the above

63rd B.P.S.C. (Pre) 2017

Ans. (d)

Bromine is a member of halogen group of elements. It is a fuming red-brown liquid at room temperature that evaporates readily to form a similarly coloured gas.

31. Which one of the following elements exist in liquid state at room temperature?

- (a) Hydrogen (b) Oxygen
(c) Bromine (d) Iodine

U.P. P.C.S. (Mains) 2017

Ans. (c)

Bromine is an element with symbol Br and atomic no. 35. It is the third lightest halogen and is a fuming liquid with deep red colour. It is the only non-metallic element to exist in the liquid state at room temperature.

32. The halogen being used as analgesic is –

- (a) Chlorine (b) Bromine
(c) Iodine (d) Fluorine

R.A.S./R.T.S.(Pre) 2012

Ans. (b)

Analgesic drugs act in various ways on the peripheral and central nervous system. Compounds of bromine are used as analgesics.

33. Which one of the following non-metals is not a poor conductor of electricity?

- (a) Sulphur (b) Selenium
(c) Bromine (d) Phosphorus

I.A.S. (Pre) 2007

Ans. (b)